

Mohamed Dhia BEN HMIDA

Software Engineer | AI / ML

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Portfolio: <https://mdbh.netlify.app>

GitHub: <https://github.com/dhiaBenHmida>

PROFESSIONAL SUMMARY

Software engineer with 8 years of engineering experience delivering production AI / ML systems (computer vision, transformers, LLMs) packaged as full-stack products with Python / FastAPI, React, and Spring Boot. Builds training pipelines, model registries, and GPU-backed inference APIs with RBAC and operational dashboards. Also leads delivery of regulated Java / React systems in banking environments since March 2026, applying the same discipline around data quality, security, and observability.

SKILLS

Machine learning: PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, Vision Transformers (ViT), EfficientNet, ConvNext, TIMM, LoRA, TensorFlow.js, TensorBoard

Computer vision and LLMs: OpenCV, MediaPipe, large language models (LLMs), GPU-accelerated inference, 4-bit NF4 quantization, torch.compile

Backend APIs: Python, FastAPI, Java 21, Java, Spring Boot, RESTful APIs, JWT, Docker, Docker Compose, microservices

Frontend: React, TypeScript, Material UI, Ant Design, PWA

Databases: PostgreSQL, MySQL, MongoDB, Redis

MLOps patterns: model registry, versioning, rollback, A/B testing, unified inference adapters, GitLab CI/CD, GPU production deploy

Tools: Git, Docker, Docker Compose, GitLab CI/CD, Nginx, OVHcloud, Agile Scrum

Languages (spoken): Arabic (native), French (fluent), English (fluent)

PROJECTS

Independent digital health platform

Independent startup product (not an employer) | 2024 to present | PyTorch, Hugging Face Transformers, LoRA, FastAPI, PostgreSQL, Docker Compose, GitLab CI/CD

- Built a modular dermatology analysis pipeline: specialized analyzers, rule-based diagnostic matrix (ICD-10), LoRA fine-tuning of a vision-language model, then prompt and analyzer transfer into production GPU inference (Transformers, PyTorch, 4-bit NF4, torch.compile).
- ISIC 2019 (25,331 images, 8 classes): F1 0.87 standalone, 0.94 in pipeline; 97% Top-3 on that campaign.
- Shipped clinical questionnaire trees with a final LLM synthesis step, plus a treatment-safety layer (age, pregnancy, allergies, medications) before recommendations.

- Ran GPU inference in production on OVHcloud behind Nginx and set up GitLab CI/CD with SSH deploy hooks.

DermaDetectAI

Related public GitHub | React, TypeScript, Python, FastAPI, PostgreSQL, PyTorch, TensorFlow, Hugging Face Transformers | <https://github.com/dhiaBenHmida/DermaDetectAI>

- Public GitHub companion: ViT feature extraction, custom-trained MLP classifier, MedGemma conversational diagnosis, RBAC, GPU inference. Training repo: <https://github.com/dhiaBenHmida/DermaDetectAI-Training>

VisionForge AI - Centralized Computer Vision Framework

React, TypeScript, Spring Boot, Python, FastAPI, PyTorch, ViT, EfficientNet, ConvNext, MediaPipe, OpenCV, Docker | <https://github.com/dhiaBenHmida/VisionForge-AI>

- Extensible CV training and deployment platform with unified inference interface (adapters), model registry, versioning, rollback, and A/B testing.
- MediaPipe quality validation, multi-model training pipeline, hybrid Spring Boot and FastAPI microservices.

SkinSmart - Skincare Recommendation System

React, TypeScript, Material-UI, TensorFlow.js, PHP, MySQL | <https://github.com/dhiaBenHmida/SkinSmart>

- Webcam skin analysis in-browser (TensorFlow.js), conditional survey branching, personalized recommendations, and admin A/B testing.

PROFESSIONAL EXPERIENCE

Senior Technical Consultant / Tech Lead

ITSS Global | Client missions: Algeria and Congo-Brazzaville | March 2026 - Present

- Shipped 2 production Java / React / PostgreSQL systems for Banque Nationale de l'Habitat (BNH, Algeria), Nostro account matching for treasury and BPM document routing for trade dossiers, by leading end-to-end delivery in a regulated environment.
- Enabled central-bank reporting UIs and customer-document printing for bank clients by directing Bronze / Silver / Gold lakehouse pipelines and bringing Temenos Transact Document Output plus T24 enquiry go-live for a CEMAC commercial bank (Congo-Brazzaville), including 11 printed document types.

Java and Oracle PL/SQL Developer

BFI Groupe | Tunisia | July 2023 - March 2026

- Delivered Java / Oracle banking platforms (BIAT accounting centralization; Maghreb XBRL / XML taxonomies to Bank Al-Maghrib and Banque d'Algérie (COREP-style prudential and FINREP-style financial reporting); UEMOA prudential reporting) by implementing integrations, controls, and statement generation with Oracle performance tuning.
- Documented APIs and architecture while hardening high-volume transactional databases through packages, jobs, and execution-plan tuning.

Software Development Engineer

SEGULA Technologies for Groupe Stellantis | France | October 2018 - June 2023

- Built calibration, automated reporting, and data platforms for Groupe Stellantis over a 5-year engagement by delivering VBA / Excel / Access tooling and migrating persistence to PostgreSQL.

EDUCATION

National Engineering Degree - École Nationale Supérieure d'Ingénieurs de Tunis (ENSIT), Tunis, Tunisia - Graduated June 2018 (September 2015 to June 2018)

Preparatory Classes for Engineering Studies (MP track) - Institut préparatoire aux études d'ingénieurs d'El Manar (IPEIEM), Tunis, Tunisia - 2013 to 2015

Tunisian Baccalaureate, Mathematics specialty - Graduated 2013

CERTIFICATIONS

Accelerated VBA Training - Stellantis

Technical Center, Carrières-sous-Poissy | August 2019

Massinissa Framework Training - BFI

Four-month Java / MVC framework program